

Electricity (Wiring) Regulations**(Cap. 406 sub. leg. E)****Contents**

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Electricity (Wiring) Regulations

(Cap. 406, section 59)

(Enacting provision omitted—E.R. 1 of 2022)

[1 June 1992] *L.N. 100 of 1992*

(Format changes—E.R. 1 of 2022)

1. Citation

(E.R. 1 of 2022)

- (1) These regulations may be cited as the Electricity (Wiring) Regulations.
- (2) *(Omitted as spent—E.R. 1 of 2022)*

2. Interpretation

In these regulations, unless the context otherwise requires—

approved loading (允許負載量) means the maximum current demand approved by an electricity supplier in respect of a fixed electrical installation;

breaking capacity (斷流容量) means the value of current at which a switch, a switch-disconnector, a circuit breaker or a fuse is capable of breaking a circuit at a voltage and under conditions of use and behaviour specified by its manufacturer;

category 1 circuit (第1類電路) means a circuit that operates at low voltage, but does not include a category 3 circuit;

category 2 circuit (第2類電路) means a circuit for telecommunications, radio, telephone, sound distribution, intruder alarm, bell and call, or data transmission and that is supplied with electricity from a safety source, but does not include a category 3 circuit; *(36 of 2000 s. 28)*

category 3 circuit (第3類電路) means a circuit for emergency lighting, air pressurisation systems and fire services installations including fire detection and alarm, fire pumps, fireman's lifts and smoke extraction;

category 4 circuit (第4類電路) means a high voltage circuit;

circuit (電路) means an assembly of electrically connected electrical equipment supplied from the same origin and protected against overcurrent by the same protective device or devices;

conductor (導體) means a wire, cable or other form of metal used for conveying electric current from one piece of electrical equipment to another or to earth;

current-carrying capacity (載流量) means the maximum current that can be carried by a conductor as determined by using the methods in the Code of Practice for the Electricity (Wiring) Regulations, as published by the Director and as amended from time to time, without the conductor exceeding the permissible limit of steady state temperature for the type of insulation concerned as specified in the Code; (31 of 2025 s. 109)

distribution board (配電箱) means an installed group of fuses, switches, circuit breakers or other outlet points arranged for the distribution of electricity;

earth (地) means the conductive mass of the earth whose electric potential at any point is conventionally taken as zero;

earth electrode (接地極) means a conductor or group of conductors in intimate contact with earth and providing an electrical connection to earth;

earth fault current (電路的接地故障電流) of a circuit means an unintended current that flows in a circuit when a live part of the circuit comes into direct contact with earth;

earth leakage current (對地漏電電流) means a current that flows to earth, or to extraneous conductive parts, in a circuit that is not intended to be connected to earth during normal operation;

earthing (接地) means connecting a conductor with earth;

exposed conductive part (外露非帶電金屬部分) means a conductive part of electrical equipment that can be touched and that is not a live part but which may become live under faulty conditions;

extraneous conductive part (非電氣裝置金屬部分) means a conductive part liable to introduce a potential, generally earth potential, that does not form part of an electrical installation;

final circuit (最終電路) means a circuit connected directly to current-using equipment, a socket outlet or other outlet point;

isolation (隔離) means cutting off an electrical installation, a circuit or an item of electrical equipment from every source of electricity;

linked circuit breaker (連動斷路器), **linked isolator** (連動隔離器) or **linked switch** (連動開關掣) means a circuit breaker, isolator or switch the contacts of which are so arranged as to make or break all poles simultaneously or in a definite sequence;

live part (帶電部分) means a conductor that is intended to be energised in normal use, and includes a neutral conductor;

main earthing terminal (總接地終端) means the terminal or bar provided for the connection of protective conductors, including equipotential bonding conductors, and conductors for functional earthing if any, to the means of earthing;

neutral conductor (中性導體) means a conductor connected to the neutral point of a system and contributing to the transmission of electricity;

overhead line (架空電纜) means a conductor that is placed above ground and is suspended in the open air;

phase conductor (相導體) means a conductor of an alternating current circuit for the transmission of electricity, but does not include a neutral conductor;

residual current device (電流式漏電斷路器) means a device or association of devices intended to cause the opening of contacts when the residual current attains a specified amperage under conditions specified by its manufacturer;

safety source (安全電源) means—

- (a) a double-insulated safety isolating transformer with its secondary winding being isolated from earth, and having a nominal output voltage not exceeding 55V;
- (b) a source of electricity providing a degree of safety equivalent to that of the safety isolating transformer referred to in paragraph (a) (for example, a motor-generator with windings providing equivalent isolation); or
- (c) a source providing electricity at a voltage not exceeding extra low voltage and independent of a higher voltage circuit;

substation (電力分站) means premises or an enclosed part of premises that contains electrical equipment for transforming or converting energy to or from high voltage (other than transforming or converting energy solely for the operation of switching devices or instruments) or for switching, controlling or regulating energy at high voltage and that are large enough to admit the entrance of a person after the electrical equipment is in position;

switching (開關) means making or breaking a current under normal circuit conditions;

switchroom (開關房) means premises or an enclosed part of premises that contains electrical equipment for switching, controlling or regulating electricity at low voltage and above and that are large enough to admit the entrance of a person after the electrical equipment is in position.

3. Application

- (1) These regulations apply to fixed electrical installations.
- (2) These regulations, except regulation 5(1), do not apply to a category 2 circuit.

4. General safety requirements

- (1) Good workmanship and suitable materials shall be used.
- (2) A fixed electrical installation shall be designed, constructed, installed and protected so as to prevent danger.
- (3) Electrical equipment shall be installed so that it is capable of being identified, maintained, inspected and tested, where appropriate, so as to prevent danger.
- (4) Electrical equipment that requires operation, maintenance or attention shall be installed so as to provide adequate and safe means of access and working space.
- (5) The owner and person in control of a substation or switchroom shall ensure that unauthorized entry into his substation or switchroom is prevented.
- (6) The owner and person in control of premises in which a switchroom or substation is located shall ensure that their entrances and exits are kept clear of obstructions that could prevent easy access to the switchroom or substation.
- (7) A registered electrical worker shall ensure that safety precautions are taken to prevent danger arising from electrical wiring work done by him or under his supervision.

5. Segregation of circuit categories

- (1) Category 1, 2, 3 and 4 circuits shall be segregated from one another as far as is necessary to prevent danger and electrical interference.
- (2) Category 4 circuits shall not be installed in the same conduit, ducting or trunking as circuits of other categories and shall be segregated from circuits of other categories at a distance sufficient to prevent danger and electrical interference with the circuits of those other categories.

6. Circuit arrangement

- (1) A wiring installation shall be divided into as many circuits as are necessary in order to—
 - (a) avoid danger in the event of a fault; and
 - (b) facilitate safe operation, inspection, testing and maintenance.
- (2) A separate circuit shall be provided for each part of a wiring installation that needs to be separately controlled.
- (3) Where a wiring installation comprises more than one final circuit—
 - (a) each final circuit shall be separately protected against overcurrent and connected to a separate isolation device in a distribution board;
 - (b) the wiring of each final circuit shall be electrically separated from that of every other final circuit, so as to prevent indirect energisation of the final circuit intended to be isolated; and
 - (c) each final circuit shall be designed so that it is not affected by the failure of any other final circuit.

7. Current demand

- (1) A circuit shall be suitable for the maximum current demand of all current-using equipment that is connected or intended to be connected to the circuit when the current-using equipment is functioning in its intended manner.
- (2) The maximum current demand of a fixed electrical installation shall not exceed the approved loading.

8. Isolation and switching

- (1) Effective means, readily accessible for operation, shall be provided so that all voltage may be cut off from every fixed installation, from each of its circuits and from all electrical equipment connected to each circuit, as may be necessary to prevent or remove danger.
- (2) Where motorised drives are involved with moving parts that are readily accessible and that may cause bodily injury, there shall be an effective means of disconnection locally placed for ready operation so that all voltage may be cut off from the motorised drives.

9. Overcurrent protective devices

A circuit shall be protected against overcurrent by a device that—

- (a) operates automatically to break the circuit before the safe current rating of the circuit is exceeded;
- (b) is of adequate breaking capacity;
- (c) is suitably located; and
- (d) is properly constructed,

so as to prevent danger from overheating, arcing or the scattering of hot particles when the device comes into operation and so as to permit restoration of the supply without danger.

10. Neutral conductor protective devices, isolators and switches

- (1) A single-pole switch shall be inserted in the phase conductor of a single phase circuit only.
- (2) No fuse and no circuit breaker, isolator or switch, other than a linked circuit breaker, linked isolator or linked switch, shall be inserted in a neutral conductor.
- (3) Notwithstanding subregulations (1) and (2), a single-pole switch may be inserted between the earthing connection and the neutral terminals of generators running or likely to be running in parallel so long as adequate precautions have been taken to avoid danger.
- (4) A solid link may only be inserted in a neutral conductor so long as it cannot be disconnected without first disconnecting the related phase conductors or without using a tool.
(L.N. 323 of 1990)
- (5) A linked circuit breaker, a linked isolator or a linked switch inserted in a neutral conductor shall be arranged to break also the related phase conductors.

11. Earth leakage and earth fault currents

- (1) A circuit shall be provided with effective means to prevent the persistence of earth fault currents and dangerous earth leakage currents.
- (2) Where metalwork of electrical equipment, other than its current-carrying conductors, may become charged with electricity in a manner that would cause danger if the insulation of a conductor becomes defective or if a fault occurs in the equipment, the metalwork shall be earthed in a manner that will cause safe discharge of electricity, or other effective means shall be provided to prevent the occurrence of danger.

- (3) Where metalwork is earthed in accordance with subregulation (2), the circuits concerned shall be protected against the persistence of earth fault current by—
 - (a) an overcurrent protective device in accordance with regulation 9; or
 - (b) a residual current device or an equally effective device, to automatically disconnect the faulty circuit in time to avoid danger.
- (4) Where metalwork of electrical equipment is earthed in accordance with subregulation (2) and the metalwork is accessible simultaneously to a substantial extraneous conductive part, the extraneous conductive part shall be effectively connected to the main earthing terminal of the fixed electrical installation.
- (5) Subregulation (4) applies only to—
 - (a) a fixed electrical installation that was completed on or after 1 January 1985; and
 - (b) a fixed electrical installation that was substantially altered on or after 1 January 1985 or that is substantially altered after the commencement* of this regulation.

Editorial Note:

* Commencement date: 1 June 1992.

12. Earthing arrangement

- (1) A fixed electrical installation shall have an effective earthing arrangement, including a connection to earth electrodes, for its protection against earth fault currents and dangerous earth leakage currents.
- (2) In addition to the requirement under subregulation (1), where the supply is taken directly from an electricity supplier's

transformer within the premises in which the fixed electrical installation is situated, the main earthing terminal of the installation shall be bonded to the supplier's bonding terminal that connects to the earthed point of the transformer.

- (3) In addition to the requirement under subregulation (1), where the supply is taken from the electricity supplier's underground cable that has exposed conductive parts at the cable termination at the point of supply, the main earthing terminal of the fixed electrical installation shall be bonded to those exposed conductive parts.
- (4) Where bonding referred to in subregulations (2) and (3) is used to prevent the rise of a dangerous earth potential, the fixed electrical installation shall satisfy all the requirements of regulation 11 even if the bonding is disconnected.
- (5) Subregulations (2) to (4) apply only to a fixed electrical installation that is completed or substantially altered on or after the commencement* of this regulation.

Editorial Note:

* Commencement date: 1 June 1992.

13. Conductors, joints and connections

- (1) A conductor shall be of sufficient size and current-carrying capacity for its intended purpose.
- (2) A live conductor shall be protected against accidental direct contact by a person.
- (3) An electrical joint, connection and conductor shall be properly constructed and installed with respect to conductance, insulation, mechanical strength and protection.

14. Wiring installation enclosure

- (1) An enclosure of a wiring installation shall be properly

designed, constructed, installed, protected and maintained so as to prevent danger.

- (2) Good workmanship and suitable materials shall be used for the enclosure of a wiring installation.

15. Adverse conditions installation

Where all or part of a fixed electrical installation is—

- (a) in or likely to be in water or exposed or likely to be exposed to weather;
 - (b) exposed or likely to be exposed to high ambient temperature, corrosive atmosphere, polluting substances or other adverse hazardous conditions; or
 - (c) in surroundings susceptible to risk of fire or explosion,
- it, or the relevant part of it, shall be specially constructed, installed or protected to prevent danger arising from those adverse conditions.

16. Overhead line installations

- (1) A low voltage overhead line shall be designed, constructed, installed and protected so as to prevent danger.
- (2) The height above ground of a low voltage overhead line at any point of the span shall not be less than 5.8 m across any place accessible to vehicular traffic and 5.2 m otherwise.
- (3) No overhead line operating at high voltage shall be installed without the Director's approval.

17. Display of labels and notices

- (1) An identification label and an appropriate warning notice including the words “Danger” and “危險” shall be displayed in a conspicuous position at every entrance of a substation and switchroom.

- (2) A warning notice bearing the words “DO NOT REMOVE” and “切勿移去” shall be displayed in a conspicuous position at or near the point of connection of each earthing conductor to an earth electrode, and at or near each main bonding connection.
- (3) A warning notice bearing the words “CAUTION—EQUIPMENT UNDER REPAIR” and “小心——器具待修” or “CAUTION—MEN AT WORK” and “小心——工程進行中”, or bearing both warnings, shall be displayed in a conspicuous position at or near the electrical equipment and at the isolation device associated with the electrical equipment on which work is being carried out.
- (4) A notice bearing the words “This installation must be tested and certified by a grade (A/B/C/H/R) electrical worker before (date)” and “本裝置須於(日期)前由(A/B/C/H/R)級電業工程人員測試及發出證明書” shall be displayed in a conspicuous position at or near the main distribution board of a fixed electrical installation referred to in regulation 20.
- (5) The letters and Chinese characters in a warning notice or identification label shall be legible and large enough for easy reading.
- (6) An identification label and warning notice referred to in subregulations (1), (2) and (4) shall be durable and securely fixed in position.
- (7) A warning notice under subregulation (3) shall be reasonably durable and shall be placed or affixed so that it cannot be accidentally displaced.

18. Alterations and additions

- (1) No alteration or addition shall be made to an existing fixed electrical installation unless—

- (a) the ratings and the conditions of the existing electrical equipment of the affected parts are suitable and adequate for the altered circumstances; and
 - (b) the protection against overcurrent, earth fault currents and dangerous earth leakage currents for the affected parts is altered as necessary for the altered circumstances.
- (2) The maximum current demand of a fixed electrical installation after an alteration or addition shall not exceed the existing approved loading for the installation unless the electricity supplier approves a new loading for the installation.

19. First inspection, testing and certification

- (1) A fixed electrical installation shall, after completion (including any work completed after repair, alteration or addition) and before it is energised for use, be inspected, tested and certified by a registered electrical worker to confirm that the requirements of the Ordinance have been met.
- (2) In the case of a repair, alteration or addition to a fixed electrical installation, only the affected parts of the installation need to be inspected, tested and certified under this regulation.

20. Periodic inspection, testing and certification

- (1) An owner of a fixed electrical installation located in one of the following type of premises must have it inspected, tested and certified at least once every 12 months— *(29 of 2021 s.37)*
 - (a) a place of public entertainment as defined in the Places of Public Entertainment Ordinance (Cap. 172) other than a sea-going vessel;
 - (b) premises for the manufacturing or storing of dangerous goods described in section 6(a) of the Dangerous Goods

(Application and Exemption) Regulation 2012 (Cap. 295 sub. leg. E); and *(10 of 2005 s. 213; 29 of 2021 s.37)*

- (c) premises with a high voltage fixed electrical installation fed directly from a high voltage supply.
- (2) An owner of a low voltage fixed electrical installation that has an approved loading exceeding 200A, single or three phase, at nominal low voltage, that is in a factory or industrial undertaking, as those terms are defined in section 2 of the Factories and Industrial Undertakings Ordinance (Cap. 59), shall, unless the factory or industrial undertaking is a premises referred to in subregulation (1), have the installation inspected, tested and certified at least once every 5 years.
- (3) An owner of a low voltage fixed electrical installation that has an approved loading exceeding 100A, single or three phase, at nominal low voltage, that is in premises other than those referred to in subregulation (1) or (2), shall have the installation inspected, tested and certified at least once every 5 years.
- (4) An owner of a low voltage fixed electrical installation located in one of the following type of premises shall have it inspected, tested and certified at least once every 5 years—
 - (a) a hotel as defined in section 2 of the Hotel Proprietors Ordinance (Cap. 158);
 - (b) a private healthcare facility that is a hospital as defined in the Private Healthcare Facilities Ordinance (Cap. 633); *(34 of 2018 s. 183 and E.R. 5 of 2018)*
 - (ba) a scheduled nursing home within the meaning of the Private Healthcare Facilities Ordinance (Cap. 633) for which an exemption granted under section 128 of that Ordinance is in force; *(34 of 2018 s. 183 and E.R. 5 of 2018)*

- (bb) a nursing home within the meaning of the Residential Care Homes (Elderly Persons) Ordinance (Cap. 459); (*34 of 2018 s. 183*)
 - (c) a school as defined in section 3 of the Education Ordinance (Cap. 279);
 - (d) premises of the institutions listed in section 2 of the Education Ordinance (Cap. 279);
 - (e) a child care centre that is registered under the Child Care Services Ordinance (Cap. 243); and (*32 of 2000 s. 36*)
 - (f) premises that the Director may, by notice posted to or served on the owner, specify that he considers could cause great harm in the event of an electrical accident.
- (5) The owner shall deliver a certificate prepared under this regulation to the Director for endorsement within 2 weeks after the date of the certificate.
- (6) The owner shall submit an endorsement fee of \$695 for each certificate delivered to the Director under subregulation (5). (*L.N. 451 of 1993; L.N. 602 of 1994; L.N. 33 of 1997; L.N. 22 of 1998; L.N. 135 of 1998; L.N. 352 of 2000*)

21. Procedures for inspection, testing and certification

- (1) Tests on a fixed electrical installation shall be carried out with proper tools and testing equipment and in accordance with accepted practices so that no danger to persons or property can occur even if the circuit being tested is defective.
- (2) Results of tests and inspections shall be recorded, dated and certified by the registered electrical worker who carried out the tests and inspections.
- (3) Where a fixed electrical installation is divided into more than one part, the individual parts need not be inspected, tested

and certified by the same registered electrical worker, and a single certificate in a form specified by the Director may be issued and certified by a registered electrical worker to cover a number or all of the individual parts, as long as he has received the appropriate certificates certified by other registered electrical workers for the individual parts, and is satisfied that the certificates are completed and signed by a registered electrical worker of appropriate grade.

- (4) A registered electrical worker shall ensure that the tests and inspections covered in his certificate have been carried out within a reasonable time, normally not exceeding one month, before the date of certification and that the results of the tests and inspections are satisfactory.

22. Making and keeping of records

- (1) A registered electrical contractor shall make and keep all relevant records of electrical work carried out by his employees for the lesser of 5 years or the time since his registration as an electrical contractor.
- (2) The owner of a fixed electrical installation referred to in regulation 20 shall make available, for inspection as and when required by the Director, latest test certificates, and, where applicable, a written summary of safety precautions taken for each event of testing and maintenance work carried out on an installation referred to in regulation 20(1)(c).

23. Prescribed period for section 12(2)

For the purposes of section 12(2), 4 months is the prescribed period.

24. Offences and penalties

- (1) A person who contravenes regulation 4(5) or (6), 20 or 22(2) commits an offence and is liable to a fine at level 3.

- (2) An owner of a fixed electrical installation who fails to comply with the requirements of regulation 17(1), (2), (4), (5) or (6) commits an offence and is liable to a fine at level 3.
- (3) An owner of a fixed electrical installation who causes or knowingly allows another person to contravene regulation 18 commits an offence and is liable to a fine at level 5 on a first conviction and a fine at level 6 on a subsequent conviction for the same offence and in either case is liable to imprisonment for 6 months.

(E.R. 1 of 2022)